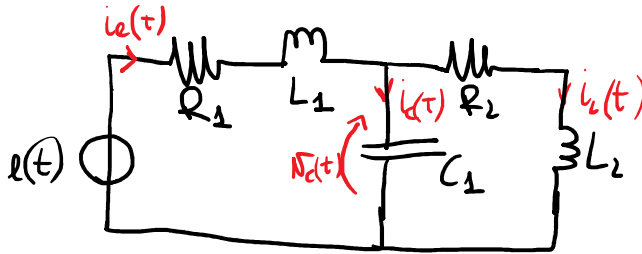


E_s. 1

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$$u(t) = \sqrt{2} \cdot 230 \cdot \sin(\omega t + 30^\circ) \text{ V}$$

$$f = 50 \text{ Hz}$$

$$R_1 = 0,1 \text{ } \Omega$$

$$R_2 = 5,2 \text{ } \Omega$$

$$L_1 = 0,64 \text{ mH}$$

$$L_2 = 12 \text{ mH}$$

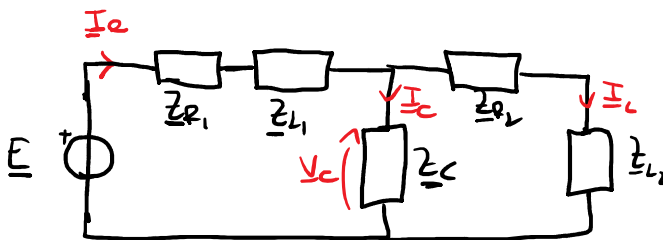
$$C_1 = 180 \text{ } \mu\text{F}$$

Calcolare:

- le correnti in ogni ramo

- $u_c(t) = ?$

① TEMPO $\rightarrow$ FASORI



$$u(t) = \sqrt{2} \cdot 230 \sin(\omega t + 30^\circ) \text{ V}$$

$$\underline{E} = 230 e^{j30^\circ}$$

$$\omega = 2\pi f = 314,16 \text{ rad/s}$$

IMPEDENZE:

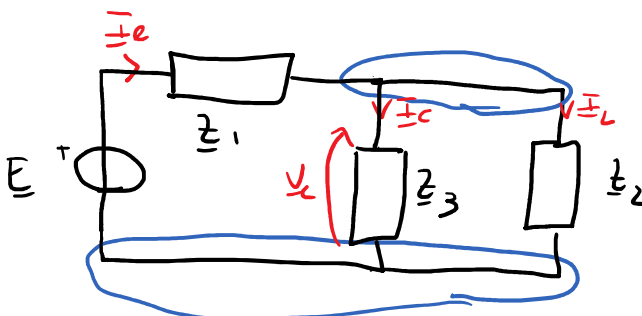
$$\underline{Z}_{R1} = R_1 = 0,1 \text{ } \Omega$$

$$\underline{Z}_{L1} = jX_{L1} = j\omega L_1 = j 314,16 \cdot 0,64 \cdot 10^{-3} = j0,2 \text{ } \Omega$$

$$\underline{Z}_C = jX_C = -\frac{j}{\omega C} = -17,66j \text{ } \Omega$$

$$\underline{Z}_{R2} = R_2 = 5,2 \text{ } \Omega$$

$$\underline{Z}_{L2} = jX_{L2} = j\omega L_2 = j 314,16 \cdot 12 \cdot 10^{-3} = 3,77j \text{ } \Omega$$



$$\underline{Z}_1 = \underline{Z}_{R1} + \underline{Z}_{L1} = 0,1 + j0,2$$

$$\underline{Z}_2 = \underline{Z}_{R2} + \underline{Z}_{L2} = 5,2 + j 3,77$$

$$\underline{Z}_3 = \underline{Z}_C = -17,66j$$



② Risolviamo il circuito nel dominio fasoriale

Applichiamo l'impedenza:
$$\underline{V}_c = \frac{\underline{E}/\underline{Z}_1}{\frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2} + \frac{1}{\underline{Z}_3}} = 197,68 + j108,58$$

$$= 225,53 \angle 28,78^\circ$$

$$\underline{I}_c = \frac{\underline{V}_c}{\underline{Z}_3} = 12,76 \angle 118,70^\circ$$

$$\underline{I}_L = \frac{\underline{V}_c}{\underline{Z}_2} = 35,11 \angle -7,16^\circ$$

$$\text{LKC: } \underline{I}_e = \underline{I}_c + \underline{I}_L = 29,49 \angle 13,34^\circ$$

③ FASORI $\rightarrow$ TEMPO

$\underline{V}_c = 225,53 \angle 28,78^\circ$

$\xrightarrow{\text{SCT}}$ $v_c(t) = 225,53 \cdot \sqrt{2} \sin(\omega t + 28,78^\circ)$

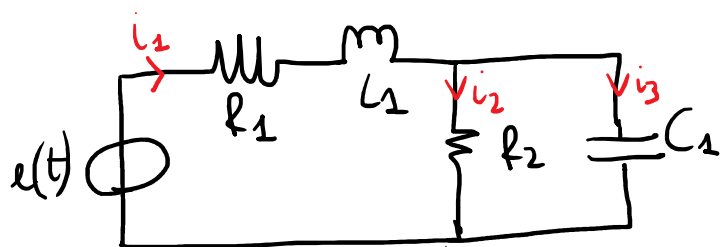
$$i_c(t) = 12,76 \cdot \sqrt{2} \sin(\omega t + 118,70^\circ)$$

$$i_L(t) = 35,11 \sqrt{2} \sin(\omega t - 7,16^\circ)$$

$$i_e(t) = 29,49 \sqrt{2} \sin(\omega t + 13,34^\circ)$$

Es. 2

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$$e(t) = 10\sqrt{2} \cos(\omega t + \pi/2) \text{ V}$$

$$R_1 = 2 \Omega$$

$$R_2 = 1 \Omega$$

$$L_1 = 1 \text{ mH}$$

$$C_1 = 500 \text{ nF}$$

$$\omega = 10^3 \text{ rad/s}$$

$$i_1(t) = ? \quad i_2(t) = ? \quad i_3(t) = ?$$

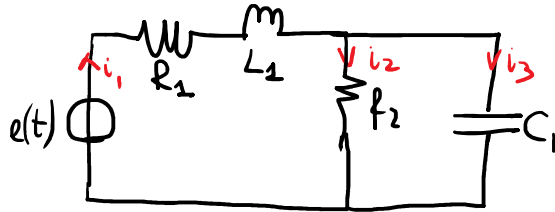
① TEMPO $\rightarrow$ FASORI

② RISOLVERE IL CIRCUITO NEL DOMINIO FASOR.

③ FASORI $\rightarrow$ TEMPO

ES.

mercoledì 4 maggio 2022 17:55



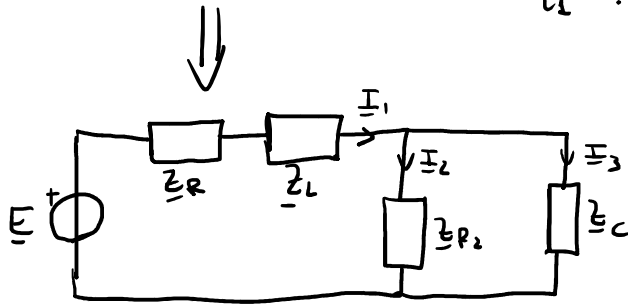
$$e(t) = \sqrt{2} \cdot 10 \cos(\omega t + \frac{\pi}{3})$$

$$R_1 = 2 \Omega \quad R_2 = 1 \Omega$$

$$L_1 = 1 \text{ mH} \quad C_1 = 500 \text{ nF}$$

$$\omega = 10^3 \text{ rad/s}$$

$$i_1 = ? \quad i_2 = ? \quad i_3 = ?$$



$$\underline{E} = 10 e^{j\frac{\pi}{3}}$$

$$\underline{Z}_R = R_1 = 2 \Omega$$

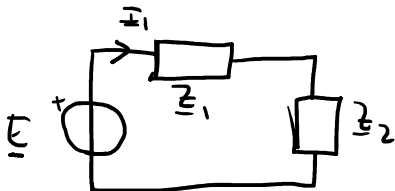
$$\underline{Z}_L = jX_L = j\omega L = j10^3 \cdot 10^{-3} = j$$

$$\underline{Z}_{R2} = R_2 = 1 \Omega$$

$$\underline{Z}_C = -j \frac{1}{\omega C} = -j \cdot \frac{1}{10^3 \cdot 500 \cdot 10^{-6}} = -j2$$

$$\underline{Z}_1 = \underline{Z}_R + \underline{Z}_L = 2 + j$$

$$\begin{aligned} \underline{Z}_2 &= \underline{Z}_{R2} \parallel \underline{Z}_C = \frac{\underline{Z}_{R2} \cdot \underline{Z}_C}{\underline{Z}_{R2} + \underline{Z}_C} = \frac{1 \cdot (-j2)}{1 - j2} = -\frac{2j}{1 - j2} \cdot \frac{(1 + j2)}{(1 + j2)} = \\ &= -2j \cdot \frac{1 + j2}{1 + 4} = -\frac{2j - 4}{5} = +\frac{4}{5} - j\frac{2}{5} = 0,8 - j0,4 \end{aligned}$$



$$\underline{I}_1 = \frac{\underline{E}}{\underline{Z}_1 + \underline{Z}_2} = \frac{\underline{E}}{2,8 + j0,6j} = \frac{10j}{2,8 + j0,6j}$$

$$= 0,73 + j3,41j = 3,49 \angle 77,9^\circ$$

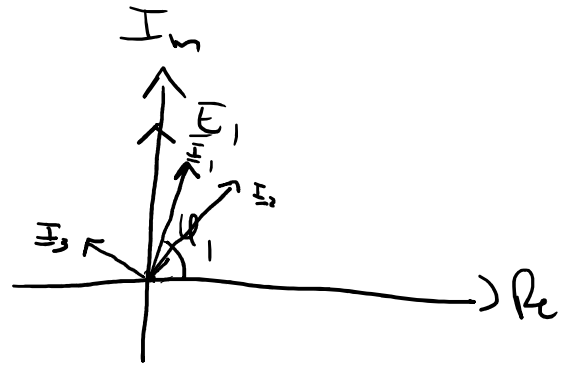
$$\underline{I}_2 = \underline{I}_1 \cdot \frac{\underline{Z}_C}{\underline{Z}_{R2} + \underline{Z}_C} = 1,95 + j2,44 = 3,12 \angle 51,34^\circ$$

$$\underline{I}_3 = \underline{I}_1 \cdot \frac{\underline{Z}_{R_2}}{\underline{Z}_{R_2} + \underline{Z}_C} = -1,22 + j0,92 = 1,56 \angle 141^\circ$$

$$i_1(t) = \sqrt{2} \cdot 3,49 \cos(\omega t + 77,9^\circ)$$

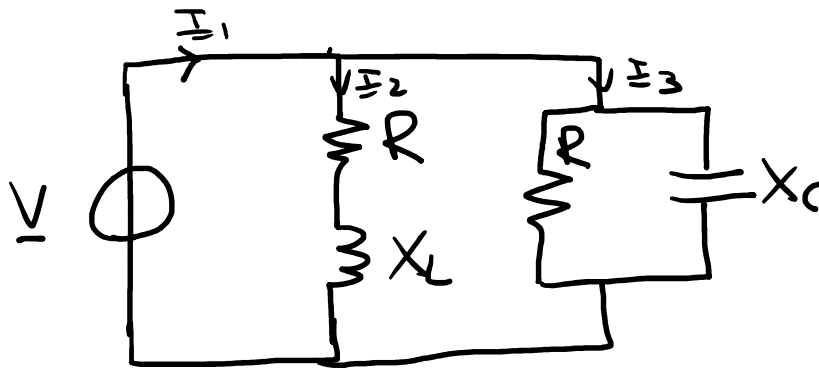
$$i_2(t) = \sqrt{2} \cdot 3,12 \cos(\omega t + 51,34^\circ)$$

$$i_3(t) = \sqrt{2} \cdot 1,56 \cos(\omega t + 141^\circ)$$



Es. 2

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$$V = 100 \text{ V}$$

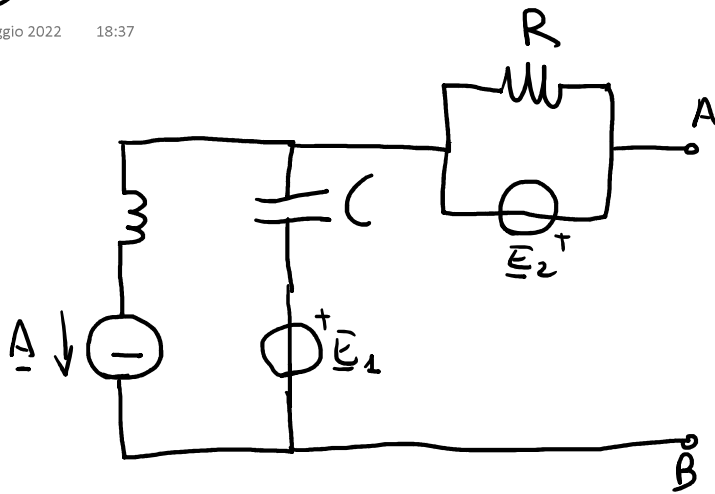
$$R = 10 \Omega$$

$$X_L = 10 \Omega$$

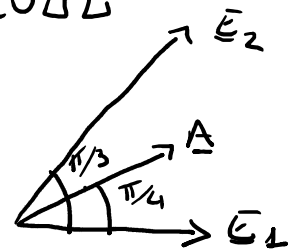
$$X_C = -10 \Omega$$

E_s.3

mercoledì 4 maggio 2022 18:37



$$\begin{aligned} E_1 &= 5V & E_2 &= 10V \\ A &= 3A & g &= 50 \text{ H}_L \\ L &= 10 \text{ mH} & C &= 20 \text{ mF} \\ R &= 10 \Omega \end{aligned}$$

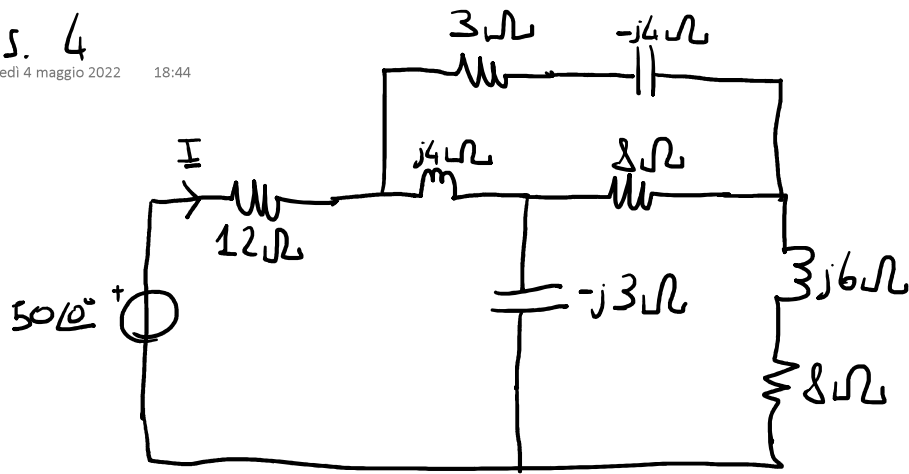


Calcolare il bipolo equiv. di Thévenin visto ai morsetti A-B e il valore del generatore equivalente $t^* = T/4$

Es. 4

mercoledì 4 maggio 2022

18:44

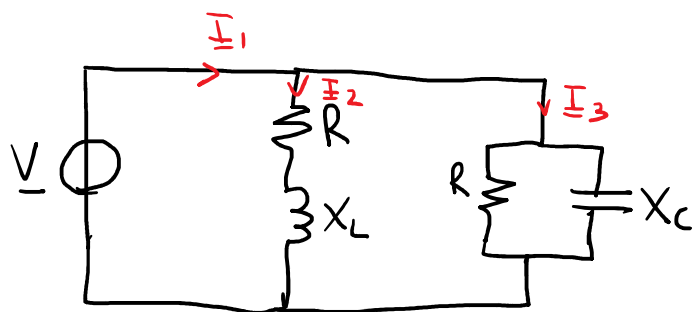


$$I = ?$$

$$\text{Sol. : } I = 3,66 e^{-j4,2^\circ} \text{ A}$$

Es. 1

mercoledì 11 maggio 2022 17:00



$$V = 100 \text{ V}$$

$$R = 10 \Omega$$

$$X_L = 10 \Omega$$

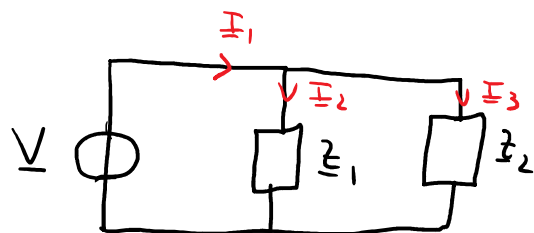
$$X_C = -10 \Omega$$

$$I_1 = ? \quad I_2 = ? \quad I_3 = ?$$

$$V = 100 e^{j0}$$

$$Z_1 = j10 + 10 \Omega$$

$$Z_2 = \frac{-10j \cdot 10}{-10j + 10} = \frac{-100j}{10 - 10j} = 5 - 5j$$

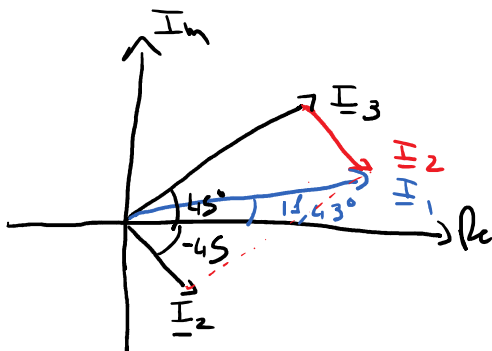


$$I_2 = \frac{V}{Z_1} = \frac{100}{10 + 10j} e^{-j45^\circ} = 5 - 5j \text{ A}$$

$$I_3 = 10 + 10j = 14,14 \angle 45^\circ \text{ A}$$

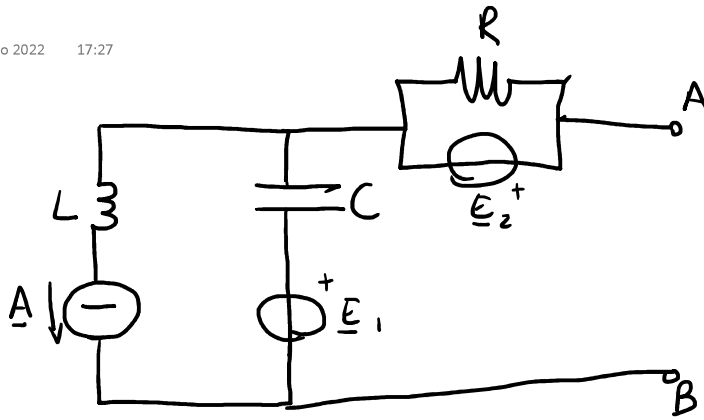
$$I_1 = I_2 + I_3 = (10 + 5) + j(10 - 5) = 15 + 5j = 15,81 \angle 18,43^\circ \text{ A}$$

Rappresentare questi fasori sul piano complesso:



Ej. 2

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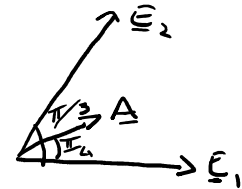


$$E_1 = 5 \text{ V} \quad E_2 = 10 \text{ V}$$

$$A = 3 \text{ A} \quad f = 50 \text{ Hz}$$

$$L = 10 \text{ mH} \quad C = 20 \text{ mF}$$

$$R = 10 \Omega$$



Bipolo eq. di Thévenin

$$E_1 = 5 \text{ V}$$

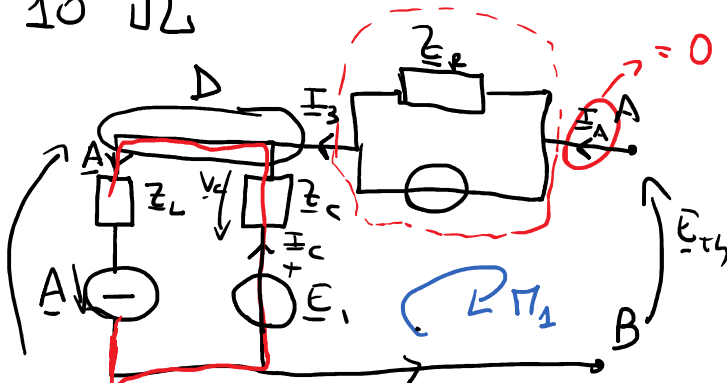
$$E_2 = 10 e^{j\pi/3} \text{ V}$$

$$A = 3 e^{j\pi/6} \text{ A}$$

$$Z_L = j\pi \Omega \rightarrow Z_L = jX_L = j\omega L = j2\pi \cdot 50 \cdot 10 \cdot 10^{-3} = j\pi \cdot 1000 \cdot 10^{-3}$$

$$Z_C = -\frac{j}{2\pi} \Omega$$

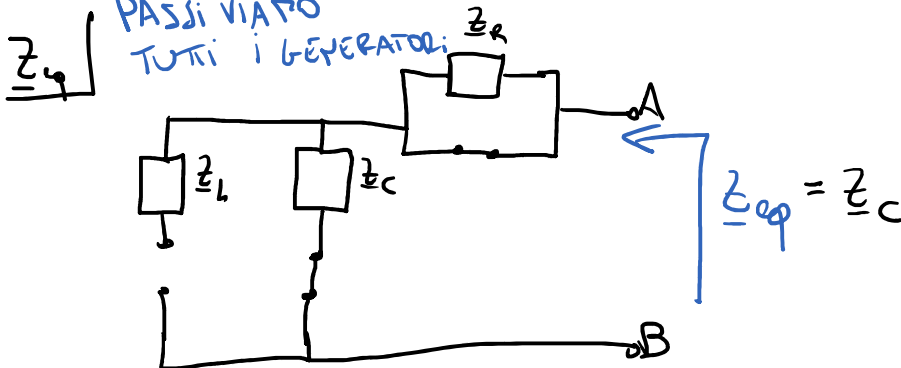
$$Z_R = 10 \Omega$$



$$I_A = I_3$$

$$\hookrightarrow I_3 = 0$$

PASSI VIATO
TUTTI I GENERATORI



$$\boxed{\underline{E}_{TH}} \quad \underline{E}_1 - \underline{V}_C + \underline{E}_2 - \underline{E}_{TH} = 0$$

$$\begin{cases} \underline{E}_1 - \underline{Z}_C \underline{I}_C + \underline{E}_2 - \underline{E}_{TH} = 0 & \text{LKT } \Pi_1 \\ \underline{I}_3 + \underline{I}_C - \underline{A} = 0 & \text{LKC } D \end{cases}$$

$$\underline{I}_3 = 0 \Rightarrow \underline{I}_C = \underline{A}$$

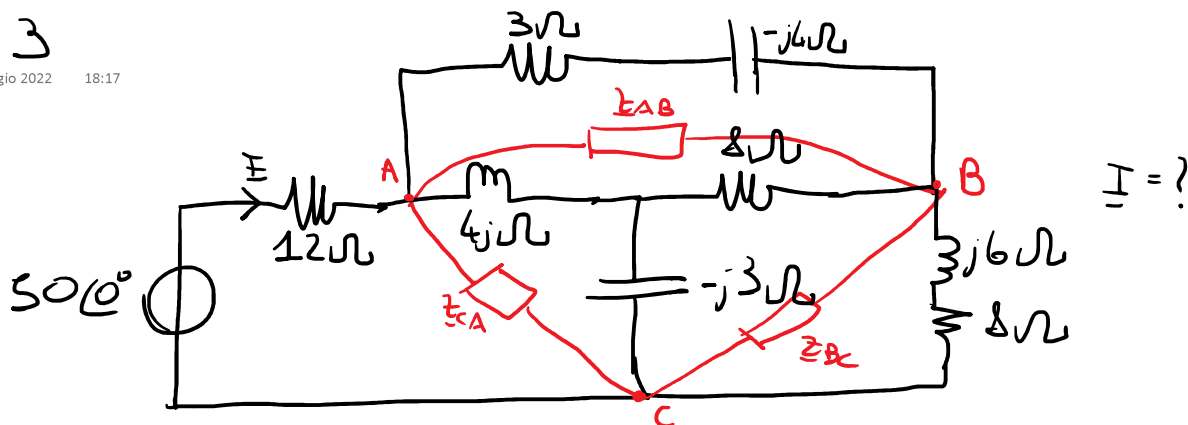
$$\underline{E}_{TH} = \underline{E}_1 + \underline{E}_2 - \underline{Z}_C \underline{A} = 13,20 \angle 43^\circ \text{ V}$$

$$e_{th}(t) = 13,20\sqrt{2} \cos(\omega t + 43^\circ) \text{ V}$$

$$\begin{aligned} t^* = \frac{T}{4} \rightarrow e_{th}(t^*) &= 13,2 \sqrt{2} \cos\left(\frac{2\pi}{T} \cdot \frac{T}{4} + 43^\circ\right) \text{ V} \\ &= 13,2 \sqrt{2} \cos(90^\circ + 43^\circ) \text{ V} = 13,2 \sqrt{2} \cos(133^\circ) \\ &= -12,73 \text{ V} \end{aligned}$$

Es. 3

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$$Z_1 = 3 - 4j \Omega$$

$$Z_2 = 8 + 6j \Omega$$

$$Z_{AB} = \frac{Z_A \cdot Z_B + Z_B \cdot Z_C + Z_C \cdot Z_A}{Z_C}$$

$$Z_{BC} = \frac{Z_A \cdot Z_B + Z_B \cdot Z_C + Z_C \cdot Z_A}{Z_A}$$

$$Z_{CA} = \frac{Z_A \cdot Z_B + Z_B \cdot Z_C + Z_C \cdot Z_A}{Z_B}$$

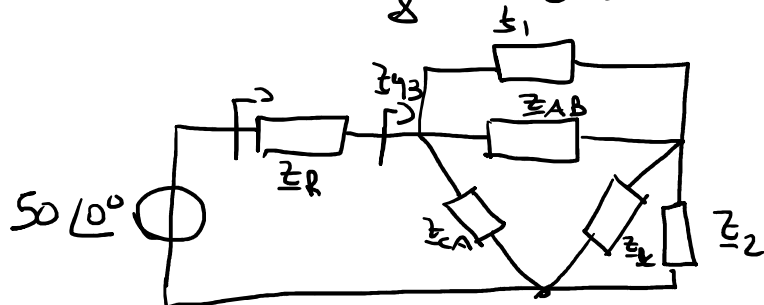
$$Z_A \cdot Z_B + Z_B \cdot Z_C + Z_C \cdot Z_A = 4j \cdot 8 + 8(-3j) + (4j)(-3j) =$$

$$= 32j - 24j + 12 = 12 + 8j$$

$$Z_{AB} = \frac{12 + 8j}{-3j} = \frac{12j - 8}{3} = -\frac{8}{3} + 4j \Omega$$

$$Z_{BC} = \frac{12 + 8j}{4j} = \frac{12j - 8}{-4} = 2 - 3j \Omega$$

$$Z_{CA} = \frac{12 + 8j}{8} = \frac{3}{2} + j$$



$$Z_{BC} // Z_2$$

$$Z_{AB} // Z_1$$

$$Z_{BC} // Z_2 = \frac{(8 + 6j)(2 - 3j)}{10 + 3j} = 2,19 - 2,04j \Omega$$

$$Z_{AB} // Z_1 = \frac{(3-4j)(-\frac{8}{3}+4j)}{3-4j-\frac{8}{3}+4j} = 24+68j \text{ } \Omega$$

$$Z_{eq2} = Z_{BC} // Z_2 + Z_{AB} // Z_1 = 26,79+65,96j \text{ } \Omega$$

$$Z_{eq3} = Z_{CA} // Z_{eq2} = \frac{(26,79+65,96j)(\frac{3}{2}+j)}{26,79+65,96j+\frac{3}{2}+j} = 1,46+j \text{ } \Omega$$

$$Z_{eq} = Z_{eq3} + Z_R = 13,46+j \text{ } \Omega$$

$$\underline{I} = \frac{\underline{E}}{Z_{eq}} = \frac{50 \angle 0^\circ}{13,46+j} = 3,69-0,27j \text{ } A$$